

A Packing of 67 Lines in $\mathbb{R}^{14}$

Limin Ge*

October 8, 2026

Abstract

We give 67 1-dimensional subspaces of $\mathbb{R}^{14}$ whose smallest squared chordal distance is at least 0.914710. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 0.913762, obtained by deleting 1 of the 68 lines of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 67 subspaces of dimension 1 in $\mathbb{R}^{14}$ whose pairwise squared chordal distances $D(a, b) = 1 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 0.914710 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 14$, $n = 1$. The cell $N = 67$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 0.913762, obtained by deleting 1 of the 68 lines of the packing found by this search. The Rankin simplex bound for this cell is $1(14 - 1)/14 \cdot 67/66 = 0.942641$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m14n1N67.txt`

in <https://github.com/Ge-limin/math>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 72 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:
14292345728510204593363308124147474813061078507
denominator:
15625000000008973695415728281548032567756440043

B Coordinates

67 blocks of 14 rows with 1 integers each; divide by 10^{12} . File: `grassmann/packings/m14n1N67.txt`.

-517259070392	-53251150291	-134811418973	108811563843
-158287414829	64695166754	-406020943053	235153140659
513485838277	224498889380	-287090991020	364077706316
25778573676	-164000496960	344009550733	-248456200549
-81403701579	-406244932865	-299852425756	-158039260391
-294883260026	36256116931	199680830806	-115826987348
217455884068	25735773162	-5747750593	323230293753
139170702558	407472213048	-358422352867	331288754552
-43124624215	184508669091	283051753140	276193049918
387646384764	-153115897438	16120127370	-184002266090
42356063031	-23425306988	468313519054	12682329869
356334632461	-162824035746	29811807483	285079786231
43713985975	51383348777	-883890529	522460036476
2981613961	703881355742	-239458357932	-146363838417
-215853004458	253037275382	-321290715099	449504602630
-119820166464	-232508636450	34134943259	68114751114
-54154831380	131549490156	102169612806	-64770190495
-255816064202	267788744715	-224498253789	-66730766462
274557188300	165174347897	614590276475	-423261485497
347789683229	350328911456	-110089943937	-411998833533
-246897931324	383509414966	300896639946	155633866056
51650887515	208877424893	-267608838770	-170671956539
-334950519316	12762066552	-86768064201	169763341070
-196652919134	380212489909	-59711182733	-375991512422
306798608226	-43369935600	246090928022	395953117420
407508882851	-129847760490	179379472565	-70766099515
-416255587969	-515189234902	422813507013	-16890689224
162445303127	-152603675845	13098856385	-223947454494
217908199754	-44839034279	-14186409422	-280848095528
118099783494	182822919626	224838962465	-12916427936
399688280503	126367664132	287169838141	-183545556398
500303759466	331755190334	-415222708279	207894678551
-61216576870	-67769644621	-65714507901	-106711519175
-425839083745	-258275639997	208231614123	724046045901
104852255946	-403959795664	-314873162368	-30410611096
540699485051	139302367548	-157490129268	298961717704
-106925875567	215523242201	65193444166	33511463625
-37463740375	191153016041	-246305024869	-190401080771
40068566964	-363335930194	118332951050	189691827968
-24736653014	424962390646	-386747446769	-293433834030
-157783073043	175646942392	-524179863505	-202236183026
-11313538892	397743167977	139590715464	133243902775
-270425415162	190152291914	-159596170215	42549231669
39328662797	-376549216578	349655843814	-24369996433
366262062196	249748389350	200549267206	21646091712
12401820421	-162053904988	-96568397692	218954705170
221955423570	592542456121	317117479642	-456761341383
-42316992077	200121948237	455466990563	53175274409
-414325940720	-129851552806	-30606109661	-65468006017
-670753392680	-170431829019	-52347196330	-611643415709
8276451346	448554245281	226974666755	-51475739852
-7814872578	182670970193	-566578394373	-319265822079
-73256004558	-157112894856	-267024277465	-35340298949
294903637499	-35808667645	50339492372	-245951989421
-147431907781	122263641266	184350943407	8400238199
64732325935	-144459551412	103362832436	439250942630
-150751668313	163483969352	-259635030797	220165588376
582690583564	-398316512793	326856982668	60772658183
118833383190	224013328572	450270193891	194375033920
225003914055	463659914105	-283349839265	245423714108
105646654977	-314233827101	-199627112490	-152403150871
-70846047142	261101016826	-132691587946	-167007039396
263506039544	-82830161313	227870136902	580535479
-86180780778	269876704708	-411482260498	21105537514
-125627954018	129439425022	-234810230299	609773674460
-98705457215	-365850286762	-153885989375	281887340299
-142745778840	-41108087957	-129476958767	-52965638864
632648787648	74195472406	-211237583727	181605195051
-97718732093	379925634652	145505571970	-117117042154
-155754933996	26537175169	-320388951298	-545407075584

126871193221	364927975952	-220884341614	240611745410
307078307340	-400097631854	65675817196	146678599515
-551251372215	-199166020765	-467550550668	-56116945543
291297804392	581397834	-78149240228	43627339867
141918710080	-438449650504	553881082949	-207855110881
-400072271204	-424563348224	231826470064	-36688951622
-284206173131	332962454986	-95676447578	-38672816163
260220310120	-224336897392	-423063654335	-464064733742
-213614095684	-92445247170	193877024085	458273907047
134207749604	-192021852302	450512885039	303291354068
-268406426110	19847064321	-131114557930	299734795941
-72235913222	-400283685810	-96032793816	122120367045
-72499182364	85212142880	111388038106	-414374028628
-161334299114	147816976175	-185564956959	225205488078
67843973957	-4040047557	-186589165223	-306357012113
-332671526875	-362742428481	-43335729159	102533584378
359284875495	-8197341946	-583154425178	432971166037
179845540398	-146435924380	202932671480	-138242915948
113534148832	177515299805	-275069304361	-13287227325
-282997067705	80469441516	66713204620	-75212927959
118623980728	427496181331	15186838971	307761676162
4885083061	120945982967	255885436469	-264805082562
17834043790	350231154606	-128253202341	-211983664981
43422202282	546927087659	-59169561096	15174884963
316190796793	-458243301517	264827392789	113747342831
-325485957903	168466775264	90653784774	-200869111565
435250580573	-274528033932	-109557537929	-212066408996
-467610539549	196160359187	381257709496	89078570834
98557301371	-360457927538	499869234116	-655870432670
536037730468	-319501167424	232149551623	-203039311889
63335610871	-1396841846	-239837582942	-219549216100
30852754255	-421668699156	-513445955166	32411794348
-220119014630	80236233414	-212320086461	88160554440
40729017247	331034307997	92945311363	567490308748
172074620104	-263671740026	227506797262	-259991458833
305402309127	105369388256	-43551529704	273649081740
68944915833	15259340918	608141877116	-179085689823
665454358829	-202707251010	-62249825605	155230723899
143698655072	314582781740	-37818846604	-369354646203
146003308282	14284544248	358767227566	358277467147
-118266167922	104854026179	-174062777340	72111527129
146489514923	-213585001759	198216765378	122266216609
606053901914	332320308515	-55204421097	-11209231468
-167030158065	297080628122	107076288803	-370435011567
34767303303	331030033619	-277910468849	40122501798
436273929965	-455338359480	498910985839	-17991277527
-46235431667	-366060922224	-111593719145	-126260396604
123032800758	-118288438283	-140669806041	289064859828
-243894691984	-304855317751	578176699535	503309600792
-366435410564	179418764910	-189463132721	12562225549
-41385698092	-224756705402	-191126056480	-237536050793
396558361608	70547328938	62199633154	189913994469
-124182515418	262724166772	-187704609143	29397070988
78499581731	118474398	-323805233586	-103645037034
144843276065	38793565799	93621102094	-35025539706
26214586998	195000397140	134576668867	-334922030458
236311160749	366520594333	-313432071872	610354921948
205348139765	-105619305253	-270549262196	234644595537
-397533061	-291758183023	-257916308012	-181017698373
-272064355175	-50725670175	110994378747	-54111195026
-270031097995	255364812546	98820579625	-235040384710
-67168224607	-403248718766	413634422115	206588665396
-495668474857	288718116027	190999267580	62650323908
-232050209606	536034587762	397159610417	277123905297
-611951353427	202934816572	4085163260	-720167506919
140862839616	-136245005738	207767120217	-20209188231
-172788642915	123506893716	-133632927085	298377419790
84249778369	-419278197276	-57862247787	-67895979530
-129815283708	-194806184019	-121126416589	-258481208340
-53613036470	-90820296798	-230177532836	50846244245
-220597339826	524466115683	-49176154750	14507673566
548147065821	-336986527327	291798503548	-321499679879
367402017053	299330111462	206424877044	-223761497854
61076121666	76999968838	-624371208401	-496793989734
132236257193	-294683220473	-150318647032	-154733500030
-416657343149	141232391396	-362118160604	-406428957900
32433543153	-270854036779	521009234617	-366699464318
-92589894328	-313095930188	407004674641	-288415687404
387224807192	-19623531384	28465973988	410550884856
-50274074096	-6409788674	129531914151	-91473130530
275518780117	-485629770593	517887423733	94918720179
-88316039054	-425032136911	-132532810855	-29220233536
-28100861104	-269160447250	410443600806	-72455098733
272895524651	-111773087827	15599600406	-28369637339
-293952592717	328513021537	176552989008	288036902031
85641608975	-376432509454	-44271685473	-143334857521
219287617044	-186721247677	15380002444	-459485144551
-589219804760	-51119761896	-118179866958	-83858985346
-153641552622	322882827598	71825981031	-240417736831
-117593491084	88633544149	-196601540059	-251857743233
-250556116227	114853112803	-369702114354	101807738533
5315035152	175893402632	-43948183051	-41477674953
427096657431	-224557242705	-418692312170	-273191293627
150396660927	-69322707172	134887633396	120599517226
-428746571404	55227200763	-225774328114	-371199548142
113589062531	-313475297162	36719557508	129866555421
-23934120173	361243074111	36239352031	-48934281276
-91761254292	-233186005609	213766057556	-354351157769
40197995992	-94319002683	20697431415	335475521348
273114415918	63756485423	-324942073422	-405200645131
87876633160	-157349417228	-596312877966	-462913239086
-63644808525	-194291860287	-165969816647	-156429882553

-351315782267	365582548157	420722885464	105701023447
121099273317	237645837220	-176613460101	253884163930
64745414169	376549839574	-140468236885	376862105654
196954358282	464626799348	7823181781	261082847609
595921711707	-288969446082	256659062662	-213829765779
-160411226958	133683660209	215144940979	185939634903
-204027215021	-217756591837	222471439770	206802536625
121946407264	-222178082040	-316891471085	46058897414
65572433091	-379825713790	-431936317486	-21482040767
247884420046	-157125633294	-423562841868	280399111881
-240478484382	305736044295	-153986282997	333245892003
138713424328	-19660712063	-172853387478	172739561600
129513912856	267691398625	44079007200	-345122115981
135248543548	545261985612	570733357270	146513877195
-637001971364	-330308378021	7991709260	329055600075
-371830448614	121899087666	-75185227691	183400898149
99486341088	186543551965	191927559667	-294668710051
-127530353018	-153490780970	147163629682	108569463741
265365532742	-266908765410	-213239269926	565203303791
30273561297	29858045829	-217376373801	-18759975591
-237012473330	-480864407026	22637822962	-15950080071
-279173416760	252184358474	372769582235	429023578469
52484489292	188870066720	-297178503620	-342589138701
375973261612	281879741936	29231331285	-416166478307
110848854715	-234726407376	492254189285	36724692744
-181375338805	-583176877816	-210792886213	-85078596196
-308483901810	174583736373	-194214089695	445837111418
-79386858718	-369927621496	-109852281500	-336803379461
96407500195	105105373862	285329063435	84934237047
55980051906	30671156028	124851285256	344959963189
361396997111	119657120889	186655312832	-167869222326
-90460098862	17854218952	201853507953	-11011000539
32964376727	3276438499	447536884508	-222404442562
345387595281	-26109174158	347766506512	100865425256
-159807042077	-236669694921	-525579060880	-402161106224
313836078419	191922802419	-260226104341	249268548541
520811474084	62531084333	-175288555511	-348546236143
-354853471346	82225989668	57181959941	65683945219
-197255405535	-305257037227	171421347938	-69456155233
258516013189	386328824113	-221505954398	-46770082505
379300365239	-587252607211	28008234695	-71258752330
-119098829183	300106628149	-2198241056	-13889266798
-221796278705	123005947328	458835634770	397562251982
-247616198025	23786511203	395460868078	-138050102700
-427881386243	-296260947877	-15463745151	93628975284
340904047410	-334165288899	187613860026	-414104032064
482374046	-57493232386	58005090794	520110096878
-50556499508	618686242679	-204914497231	94467315022
-181144486335	25862508014	-14408071319	109436442966
78779585064	-73456337384	-451289686521	-40876724576
493330879697	-528507739151	173115358437	-327234334832
132659050382	-138484916209	15608757720	-57948646855
-84068620988	85089279821	-369504330299	-497391336625
350034608269	131811045096	-425303014985	-250165617914
121320939306	-59274325305	170354809437	384527179168
-286905483260	126863867465	103791224316	-14989876860
3860207427	393720311939	-96382416655	-21220898729
-297202984580	-191934653565	46269368546	349995392724
-190679953156	-165267347005	200760200028	413963475869
43256248742	-57899151085	698187016158	-228376831085
-362769247416	-214293493423	294651807464	249905312672
8178716784	149461998459	412061016	
115871755219	-260823741834	51422050615	